Curcumin as a possible lead compound against hormone-independent, multidrug-resistant breast cancer.

We examine the possible evidence that the phytochemical curcumin may overcome resistance to hormonal and cytotoxic agents in breast cancer. We present our observations on MCF-7R, a multidrug-resistant (MDR) variant of the MCF-7 breast cancer cell line. In contrast to MCF-7, MCF-7R lacks aromatase and estrogen receptor alpha (ERalpha) and overexpresses the multidrug transporter ABCB1 and the products of different genes implicated in cell proliferation and survival, like c-IAP-1, NAIP, survivin, and COX-2. Nevertheless, in cytotoxicity and cell death induction assays, we found that the antitumor activity of curcumin is substantial both in MCF-7 and in MCF-7R. We elaborated the diketone system of curcumin into different analogues; the benzyloxime and the isoxazole and pyrazole heterocycles showed remarkable increases in the antitumor potency both in the parental and in the MDR MCF-7 cells. Furthermore, curcumin or, more potently, the isoxazole analogue, produced early reductions in the amounts of relevant gene transcripts that were diverse (i.e., they were relative to Bcl-2 and Bcl-X(L) in MCF-7 and the inhibitory of apoptosis proteins and COX-2 in MCF-7R) in the two cell lines. Thus, the two compounds exhibited the remarkable property of being able to modify their molecular activities according to the distinct characteristics of the parental and MDR cells. We discuss also how curcumin may (1) exert antitumor effects in breast cancer through ER-dependent and ER-independent mechanisms; and (2) act as a drug transporter-mediated MDR reversal agent. Overall, the structure of curcumin may represent the basis for the development of new, effective anticancer agents in hormone-independent MDR breast cancer.